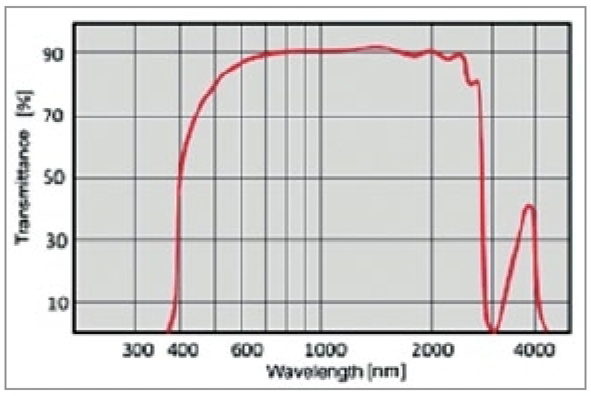

Fig. 7: Transmission versus wavelength of ZERODUR

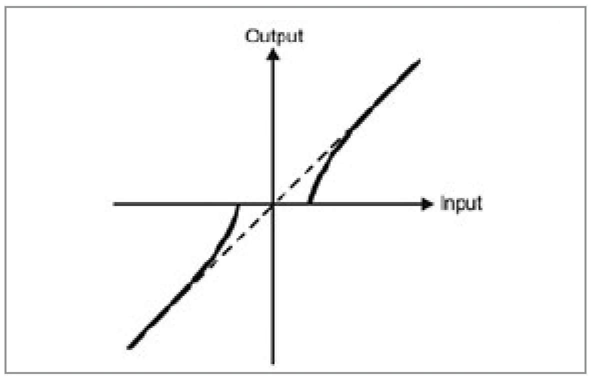

Fig. 8: Lock-in effect in RLG

low rotation rates. When the rotation rate is very low, frequencies of counter-propagating laser beams become almost identical. This leads to the two beams getting injection-locked to a common frequency. As a result, it does not respond to rotation. This is illustrated in the transfer characteristics of Fig. 8.

This problem is largely overcome by using forced dithering, in which ring laser cavity is rotated clockwise and counter-clockwise about its axis, using a mechanical spring driven at its resonance frequency. Typically, a peak dither rate of 1Arc-sec/sec is used. Dither frequency is in the range of 400Hz - 500Hz.

Parameters describing RLG's performance. Important parameters that describe RLG's performance include bias error, bias drift and instability, scale factor stability and angular random walk.

Bias error. Bias error of a rate gyroscope in general is the output produced by the gyroscope when it is not experiencing any rotation. It is defined as the voltage or percentage of full-scale output, representative of the rotational velocity in °/s as measured by the gyroscope in the absence of any rotation. Bias error is primarily caused by a number of factors, including calibration errors, bias drift, bias variation with temperature and effect of shock or g level.

Bias drift. Bias drift refers to variation of bias error over time, assuming all other factors remain constant. This is a warm-up effect usually caused by the self-heating of electrical and mechanical components of the gyroscope. This effect is more prevalent during the warm-up of the first few seconds and is non-existent after that. It should not be confused with the more colloquial term drift rate.

Bias instability. Bias instability is the fundamental parameter that defines how good or bad the gyroscope is. It is defined as the minimum point on Allan Variance curve, measured in °/hr. It represents the best bias stability achievable in a given gyroscope, with bias averaging taken at the interval defined by Allan Variance minimum.

As an example, bias stability error in the case of Honeywell gyroscope type GG1320 is specified to be ≤0.002°/hr.

Allan Variance technique is a very powerful mathematical method that allows a user to assess the real performance of any gyroscope in relation to the dynamics of its application, taking into account the effects of bias, noise, drift and long-term sensor instability.

Bias instability is always measured by averaging successive data samples. The challenge lies in selecting a suitable time over which to average the sampled data. Averages taken over short time intervals are dominated by noise, while those taken over a longer period are dominated by longer term drift. The technique involves selecting a range of time intervals, typically from 0.01s to 500s, over which to average the data. The standard deviation thus obtained from one averaged time period to the next is calculated and plotted against the averaging interval in log-log form. The resulting graph has a characteristic bathtub shape. By examining this graph, it is possible to compute the key defining characteristics of the gyroscope, namely, angular random walk, bias instability and rate random walk.

Scale factor stability. Scale factor and scale factor stability represent the transfer, or output versus input characteristics of the gyroscope. It measures the signal produced in terms of voltage or bits for a given rotation rate. It is a measure of the slope of the best straight line through the points on a graph representing the gyroscope output against input rotation rate measured over the specified dynamic range of the gyroscope.

As an example, for an analogue gyroscope, it is measured in volts per °/s. Scale factor stability and maximum rate measuring figures for Honeywell gyroscope type GG 1320 are specified to be ≤10PPM and ± 500°/s, respectively.

Angular random walk. Angular random walk, or ARW, is a measure of gyroscope noise. It is measured in °/√hr or °/√s. It can be thought of as the standard deviation due to noise while integrating the output of a stationary gyroscope over a period of time. It is inversely proportional to the square root of integration time.

As an illustration, if a gyroscope has an ARW of 1°/√s; ideally speaking, the result of angular position measurement in the ideal case would be zero. But longer the integration time, larger will be the spread of the results away from zero. The spread follows the law of being inversely proportional to the square root of integration time. In the case of Honeywell gyroscope—which is also true for typical inertial-grade gyroscopes—random walk noise is specified to be ≤0.0018°/√hr.

An INS has three orthogonally-placed RLGs to provide measurement capability of rotation in three degrees of

Fig. 9: MiniRLG2-based INS

Fig. 10: HG9900U IMU

Fig. 11: HG1700 IMU